Computer Vision & Deep Learning

Brain Tumor Detection Using Deep Learning from MRI Images

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1. Abstract

Automated classification of brain tumors through Magnetic Resonance Imaging (MRI) is one of the most important diagnostic tasks, as it provides a great opportunity to make treatment decisions. In this project, the comparative analysis of deep learning techniques for multi-class brain tumor classification based on a database of 7,023 images categorized into four classes: glioma, meningioma, pituitary tumor, and absence of any tumor, is carried out.

Five different deep learning models have been used: VGG16, ResNet50, MobileNetV2, U-Net (encoder architecture), and hybrid CNN + LSTM model. Preprocessing consisted of image resizing, normalization, and augmentation for a fair evaluation of each of the models. Accuracy, precision, recall, F1-score, confusion matrices, ROC-AUC were the metrics used for evaluating performance.

The experiment results indicate that VGG16 gave the best results, providing 97% of classification accuracy, a Macro F1-score equal to 0.97, and AUC equal to 0.9964. Furthermore, an ablation study of three parameters – learning rate, dropout rate, and fine-tuning depth was performed. For increasing the explainability of models, Grad-CAM visualization was used to highlight the important areas of an image. The research demonstrates that transfer learning based deep convolutional neural networks are a good solution for automated brain tumor classification.

2. MOTIVATION AND RATIONALE

2.1 Context and Research Theme

Image processing is one of the main application domains of computer vision and deep learning. In the context of brain imaging, Magnetic Resonance Imaging (MRI) is widely used owing to its non-invasiveness and high-resolution visualization of brain tissues. Brain tumour is a common neurological condition where timely and precise detection is essential for successful treatment planning. However, manual analysis of MRI images takes a lot of time and requires high proficiency from specialists who perform the task. Moreover, similar visual features of different tumours make it hard to diagnose them, which leads to the necessity of developing an automatic system for this task.

2.2 Problem Addressed & Significance

The purpose of this research is to design a deep learning framework to classify brain tumors from MRI images. While conventional diagnostic methods are often subjective and difficult to scale, deep learning algorithms can automatically extract discriminative features directly from the images. To explore the capability of each model, five deep learning architectures are implemented and evaluated in this study: VGG16, ResNet50, MobileNetV2, U-Net (encoder-based), and a hybrid CNN + LSTM architecture. Moreover, an ablation study is performed to examine the effect of several key hyperparameters on model performance. The significance of this study lies in providing a reproducible deep learning pipeline, together with a systematic comparison of state-of-the-art architectures for medical image classification.

3. STATE OF THE ART (SOTA)

3.1 Advancements in Image Classification with Deep Learning and Computer Vision

Recent progress in computer vision and deep learning has led to remarkable advancements in the field of image classification, which is true even in cases of medical image analysis. Classical methods of computer vision involved the development of handcrafted feature extraction algorithms based on edge detection, texture analysis, and shape descriptors. After that, traditional machine learning methods were used to analyze images and classify their contents. However, despite the success of such an approach in some cases, it had serious limitations since they failed to capture the complexity of visual patterns in medical images.

With the rise of Convolutional Neural Networks (CNN), image classification has been radically changed due to the fact that deep learning models can automatically extract hierarchical features of images without the need for prior information about their nature. Moreover, it has been shown that the technique called transfer learning, which implies training a model on large-scale datasets such as ImageNet and transferring its knowledge to other domains like medical image analysis, works perfectly well in practice.

3.2 State Of the Art Technique

3.2.1 VGG16: Classical Deep Architecture

VGG16 is among the most prominent convolutional neural network architectures and continues to be a reliable benchmark for performing image classification tasks[1]. This network is based on a simple and consistent architecture consisting of stacked 3×3 convolutional layers that allow the progressive acquisition of more complex visual features. Although having a high amount of parameters, VGG16 has shown great results in many computer vision tasks.

In the current work, VGG16 is used with the help of transfer learning with pretrained ImageNet weights. This network is used as a benchmark for testing the quality of deep feature representations in brain MRI classification.

3.2.2 ResNet50: Residual Learning for Deep Representations

The ResNet50 model pioneered the technique known as residual learning that solves the problem of vanishing gradients seen in deep neural networks[2]. This is achieved through the presence of skip connections that make the training of very deep models possible while retaining vital feature information.

In terms of the classification of brain tumors, ResNet50 can be used to learn intricate visual features from MRI scans. The deep residual nature of the model allows it to differentiate between the different types of tumors.

3.2.3 MobileNetV2: Lightweight State-of-the-Art Model

MobileNetV2 is an efficient model that is intended to be applied in environments where there is a limited amount of computing resources[3]. The model uses depthwise separable convolutions and inverted residual blocks to greatly decrease the parameter and computation count while still preserving high classification accuracy.

The current research employs MobileNetV2 in order to show how one may find a balance between computational efficiency and prediction accuracy. The model shows how lightweight deep learning models may be efficiently applied in medical image analysis tasks.

3.2.4 U-Net: Biomedical Image-Oriented Architecture

U-Net is one of the most sophisticated models that was initially built for segmentation of biomedical images[4]. The structure and skip connection used by this network allow maintaining the spatial information, making the use of U-Net particularly appropriate for medical image processing.

The initial purpose of U-Net is segmentation; however, the encoder part of the network can be used for classification purposes. In our research, we have utilized an encoder-based architecture of U-Net to extract features from the MR images.

3.2.5 CNN + LSTM: Hybrid Deep Learning Model

The Hybrid CNN + LSTM models use the benefits of convolutional neural networks and recurrent neural networks[5]. The task of finding the spatial features in the images is assigned to the CNN layer, whereas the LSTM layer learns the sequential structure in the found representations of the features.

In this research, the CNN + LSTM architecture is considered as another way of dealing with the complicated dependencies between the features found in the MRI images.

3.3 Challenges and Limitations in Existing Approaches

Although there have been many advances in the application of deep learning to the analysis of medical images, many obstacles are yet to be addressed. For example, one of the main drawbacks is the lack of quality annotated medical datasets, resulting in overfitting and decreased ability to generalize. Imbalance between classes of tumors can influence predictions made by the model as well.

Moreover, computation efficiency is an issue in case of usage of deep architectures like VGG16 and ResNet50. In addition to the heavy demands on computation resources, these types of algorithms are black box systems, meaning that it is impossible to see what happens in their "internal machinery".

3.4 Contributions of This Project

To address these challenges, this project develops a comprehensive deep learning framework for automatic brain tumor classification based on MRI images. This research work applies and compares five modern architectures reflecting different approaches in computer vision and deep learning: VGG16, ResNet50, MobileNetV2, U-Net, and CNN + LSTM.

An identical preprocessing and training pipeline is applied across all architectures to ensure a fair comparison. Moreover, the influence of architectural design on accuracy, efficiency, and generalization ability is examined. To better understand the behavior of the best-performing architecture, an ablation study is conducted by varying the learning rate, dropout rate, and fine-tuning depth. Also, Grad-CAM is used to visualize and interpret the MRI regions that have the greatest impact on the model predictions.

Overall, the main contributions of this project are a reproducible experimental framework and a comparative analysis of modern deep learning approaches for brain tumor classification.

4. OBJECTIVES

The primary objective of this project is to develop a brain tumor classification system based on deep learning methods, applied to Magnetic Resonance Imaging (MRI). This involves examining the efficiency of modern computer vision and deep learning approaches for classifying different types of brain tumors, as well as healthy scans.

To achieve this objective, five modern deep learning models — VGG16, ResNet50, MobileNetV2, U-Net (encoder-based), and CNN + LSTM — are implemented and evaluated within a single experimental framework.

A further objective is to examine the impact of architectural and hyperparameter choices through an ablation study conducted on the best-performing model. Moreover, Grad-CAM visualization is employed to increase model interpretability by highlighting the image regions most influential to the classification decision.

Finally, the project aims to establish a reproducible deep learning framework for brain tumor classification and to identify the most effective architecture for MRI-based diagnosis.

5. METHODOLOGY

The suggested approach aims at creating a complete deep learning framework that could be used for automated brain tumors classification on MRI pictures. The following aspects would be included within such a framework: data preprocessing, transfer learning, model comparison, hyperparameters optimization and explainable artificial intelligence methods in order to achieve not only the best results but also high interpretability..

5.1 Dataset and Preprocessing

5.1.1 Dataset Description

In this research, a public brain MRI dataset was used[8]. It consists of 7,023 images and includes 4 classes – Glioma, Meningioma, Pituitary Tumor, and No Tumor. In total, there are 5,712 images in the training set and 1,311 images in the test set.

5.1.2 Preprocessing Pipeline

To ensure consistency throughout all experiments, a standard preprocessing pipeline has been used.

- **Image Resizing**
  All the MRI images have been downsized to a consistent size of 128 × 128 pixels to meet the input requirement of the implemented neural networks.
- **Intensity Normalization**
  The pixel values of these images have been normalized to [0,1] for numerical stability and faster model training.
- **Label Encoding**
  The classes are converted into numeric values and trained using the sparse categorical cross entropy loss function.

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5.2 Implementation of Models

The project implements five distinct architectures to conduct a rigorous comparative analysis of diagnostic efficacy.

5.2.1 VGG16 (Classical Deep Architecture)

The implementation of VGG16 involves the use of transfer learning with pretrained weights of the ImageNet dataset[7]. The initial convolution layers are kept fixed to ensure the generic nature of the features, and the final layers are trained for classifying brain tumors. A classifier head made up of fully connected layers and dropouts is included to train the network on the MRI dataset.

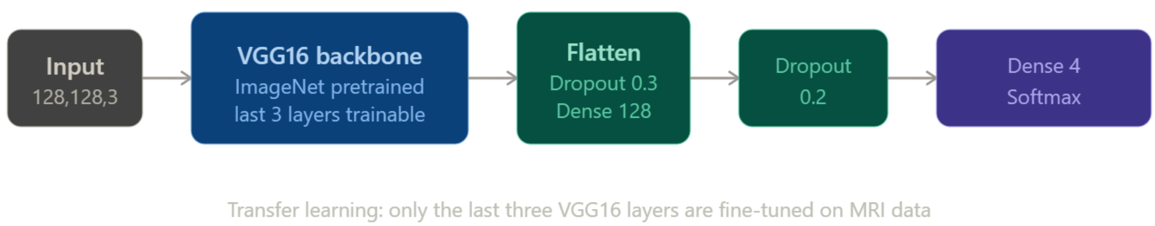


Figure 5.1: VGG16 model architecture

5.2.2 ResNet50 (Residual Learning)

ResNet50 makes use of residual learning via skip connections that facilitate training of deep architectures without falling into vanishing gradient issues. The network is fine-tuned using transfer learning and acts as a high-capacity model for MRI features.

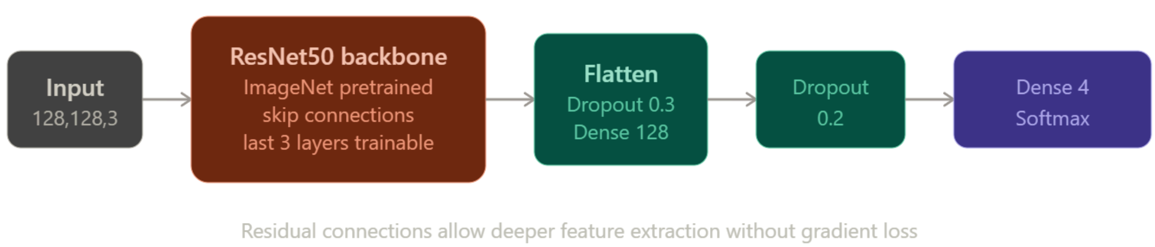


Figure 5.2: ResNet50 model architecture

5.2.3 MobileNetV2 (Efficient Design)

MobileNetV2 model is applied to serve as a smaller version compared to other larger CNN models. With the help of depthwise separable convolutions and inverted residual blocks, feature extraction and classification are done efficiently.

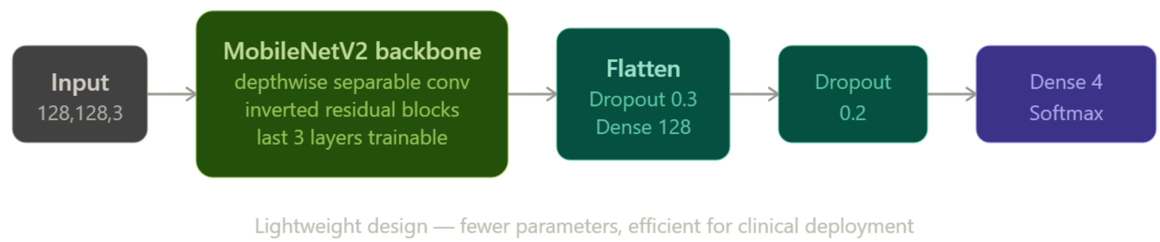


Figure 5.3: MobileNetV2 model architecture

5.2.4 U-Net (Encoder-Based Classification)

While primarily built for biomedical image segmentation, an encoder part of U-Net is modified to perform classification tasks. The encoder extracts hierarchical spatial features of MRI images, and further dense layers classify the tumors.

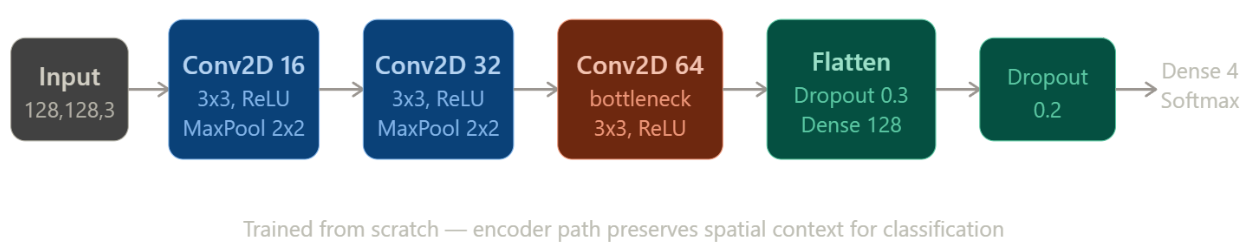


Figure 5.4: U-Net encoder model architecture

5.2.5 CNN + LSTM (Spatial-Sequential Hybrid)

The hybrid CNN + LSTM model consists of both CNN and LSTM parts. Convolutional neural networks extract spatial features from the images, and LSTM network models the features sequentially.

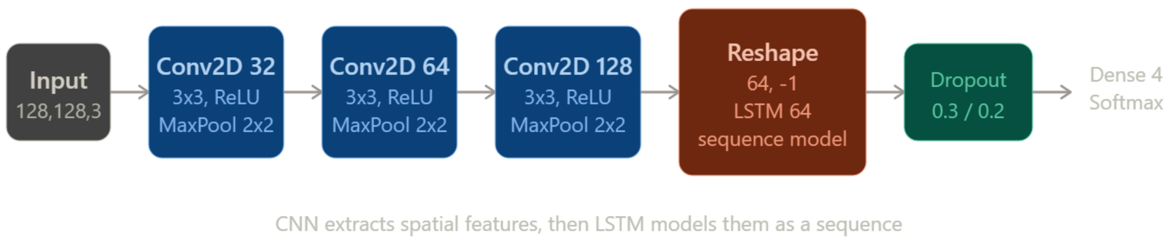


Figure 5.5: CNN + LSTM hybrid model architecture

5.3 Performance Metrics

To adequately assess the performance of the proposed deep learning models in classifying brain tumors, several performance metrics are used. In medical image analysis, relying solely on accuracy can be misleading, as different tumor classes may vary in classification difficulty. Hence, several metrics are used to evaluate both overall performance and performance per class.

- **Categorical Accuracy:** Accuracy measures the proportion of correctly classified MRI images among all test cases. It reflects the overall performance of the model and is calculated as the ratio of correct classifications to the total number of classifications made. .
- **Precision (Positive Predictive Value):** Precision is calculated as the ratio of correctly classified images for a given class to all images predicted as belonging to that class. A higher precision value indicates a lower rate of false positives.
- **Recall (Sensitivity):** Recall, or Sensitivity, refers to the proportion of actual positive samples that the classifier correctly identifies. Recall is particularly significant in medical contexts, as a missed tumor could result in delayed treatment and increased patient risk. Therefore, a high value of recall is necessary for efficient tumor detection.
- **F1-Score:** The F1-Score is the harmonic mean of Precision and Recall. This measure accounts for both false positives and false negatives simultaneously, providing a balanced evaluation of model performance. The F1-Score is especially useful when comparing classification results across different tumor types.

- **Confusion Matrix:** The Confusion Matrix displays the classification performance of a model by showing the number of correct and incorrect predictions for each class.
- **Receiver Operating Characteristic (ROC) and Area Under Curve (AUC)**
- To assess the discriminative power of the models, ROC curves and the Area Under the Curve (AUC) are computed. The ROC curve illustrates the trade-off between the true positive rate and false positive rate at various classification thresholds, while the AUC represents the model's overall discriminative ability. Higher values of AUC indicate better performance in class separation.

5.4 Feature Extraction and Visualization

A core objective of this project is to move beyond "black-box" predictions by visualizing the internal representations learned by the networks, ensuring decisions are based on pathological features rather than image noise.

- **First-Layer Feature Maps:** We implement visualization of the activations from the initial convolutional layers. These early filters highlight high-frequency components of the MRI, such as the skull boundary and the sharp edges of the tumor mass.
- **Attribution Mapping with Captum:** To validate biological relevance, we utilize the **Captum library** to generate attribution maps. These assign importance scores to each pixel, allowing us to verify if the model’s "attention" aligns with the actual tumor location identified by radiologists.
- **Feature Bottleneck Analysis:** For architectures like the **U-Net Encoder** and **ResNet50**, we analyze the bottleneck features. This allows us to observe how high-dimensional spatial data is compressed into a semantic vector representing the "signature" of a specific tumor type before reaching the final classifier.

6. EXPERIMENTS AND RESULTS

This section presents the experiments performed and the results obtained from comparing the five deep learning architectures implemented for brain tumor classification using MRI images. All the models were trained using the same preprocessing techniques and were also tested using the same testing dataset that contained 1,311 MRI images. The models were evaluated using Accuracy, Macro F1-Score, Weighted F1-Score, Confusion Matrices, ROC-AUC analysis, and classification reports.

6.1 Performance Analysis of Implemented Models

The experimental results demonstrate high diagnostic efficacy across all architectures, with performance metrics summarized below:

- **VGG16 (Best Performing Model):** VGG16 outperformed all other architectures evaluated in this experiment. By applying transfer learning with pretrained ImageNet weights and fine-tuning the last convolutional layers, the model was able to learn discriminative MRI features effectively and demonstrate strong generalization capability. The model achieved an accuracy of 96.72%, a Macro F1-Score of 0.9653, and a Weighted F1-Score of 0.9670. Very few misclassified images were observed in the confusion matrix.

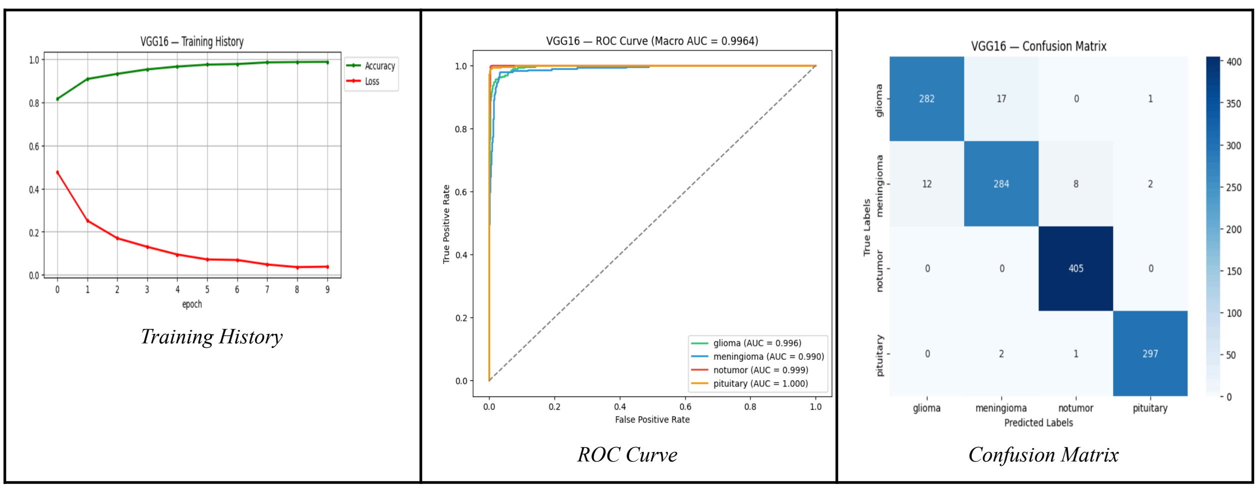


Figure 6.1: VGG16 training curves, confusion matrix, and ROC analysis.

- **MobileNetV2:** Despite its lightweight nature compared to VGG16, MobileNetV2 proved to be an efficient classifier, achieving 93.97% accuracy, a Macro F1-Score of 0.9363, and a Weighted F1-Score of 0.9396. This result clearly illustrates the efficiency of depthwise separable convolutions and the inverted residual block architecture.

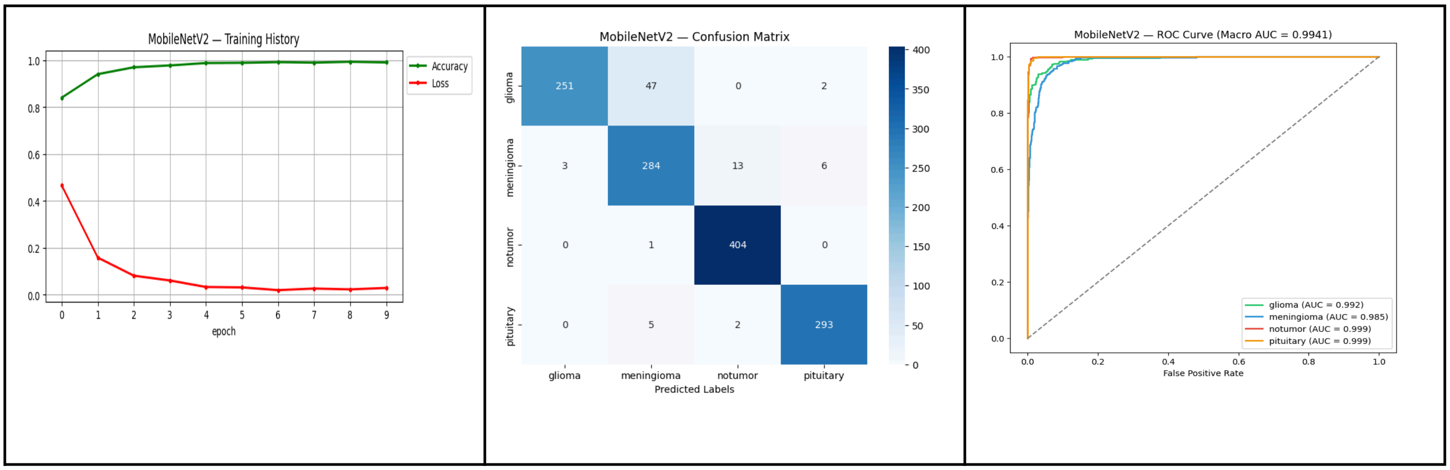


Figure 6.2: MobileNetV2 training history, confusion matrix, and ROC curve evaluation.

- **U-Net (Encoder-Based Classification):** With an accuracy of 90.85%, a Macro F1-Score of 0.9018, and a Weighted F1-Score of 0.9067, the encoder-based U-Net proved to be a strong choice for medical image classification, owing to its biomedical-oriented design and strong spatial feature extraction capability. Originally designed for segmentation tasks, the encoder was still able to extract useful MRI features for classification purposes.

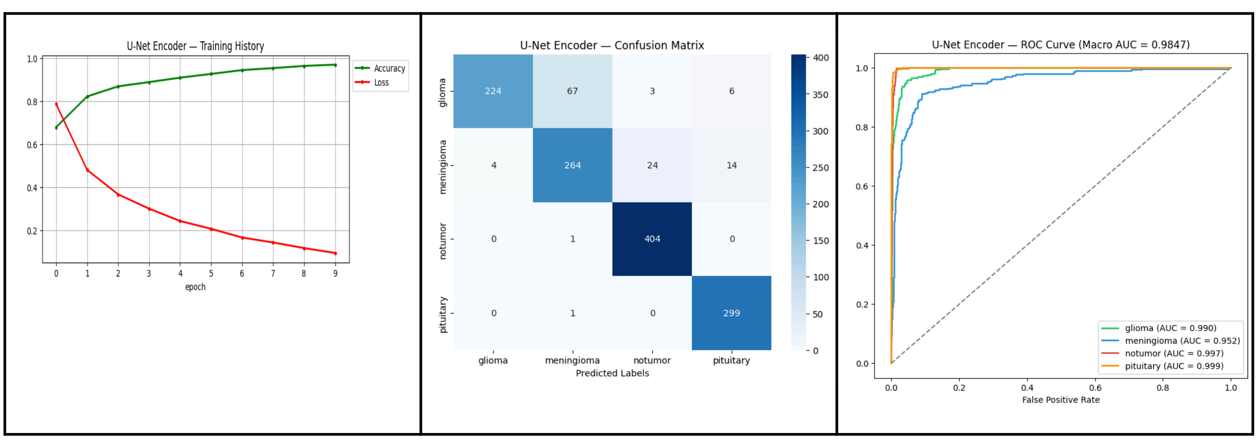


Figure 6.3: U-Net training history, confusion matrix, and ROC curve evaluation.

- **ResNet50**The ResNet50 classifier achieved 73.07% accuracy, a Macro F1-Score of 0.6726, and a Weighted F1-Score of 0.6917. Despite its residual learning mechanism, which enables very deep feature extraction, the model underperformed relative to expectations on this dataset. These results suggest that further hyperparameter tuning or longer training may be required for the model to reach its full potential on this task.

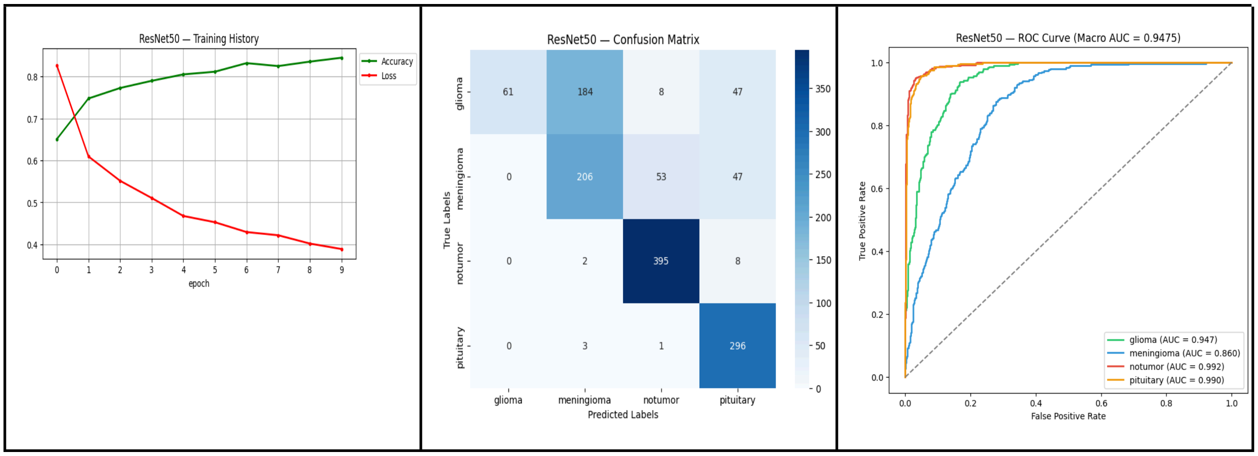


Figure 6.4: ResNet50 training history, confusion matrix, and ROC curve evaluation.

- **CNN + LSTM:** The CNN + LSTM architecture achieved an accuracy of 70.63%, a Macro F1-Score of 0.6656, and a Weighted F1-Score of 0.6814. Although the convolutional layers successfully extracted spatial features, the LSTM component did not provide any additional benefit for classifying static MRI images. The added model complexity may have negatively affected generalization performance compared to the other architectures.

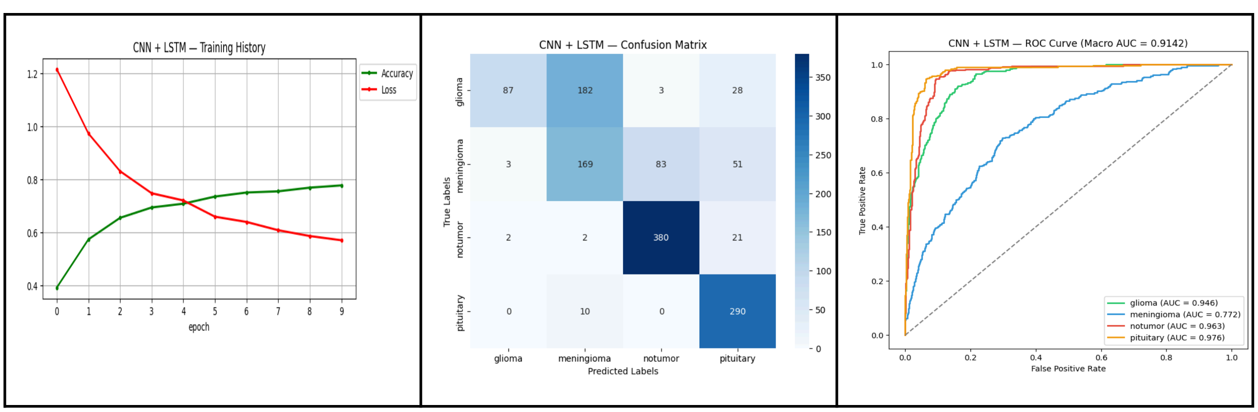


Figure 6.5: CNN + LSTM hybrid training history, confusion matrix, and ROC curve evaluation.

6.2 Comparative Summary of Results

Table 6.1: Final Comparison of All Five Models

Model	Accuracy	Macro F1	Weighted F1
VGG16	0.9672	0.9653	0.9670
MobileNetV2	0.9397	0.9363	0.9396
U-Net Encoder	0.9085	0.9018	0.9067
ResNet50	0.7307	0.6726	0.6917
CNN + LSTM	0.7063	0.6656	0.6814

The following table provides a comprehensive comparison of the classification performance across the experimental suite:

The comparative analysis has clearly proved that the VGG16 architecture has performed better than all others on every performance metric considered. The second best-performing architecture was MobileNetV2, which had substantially lower computational complexity. The U-Net had shown promising performance metrics in terms of classification along with demonstrating the significance of biomedical-specific features extraction. On the other hand, ResNet50 and CNN+LSTM showed poorer performance.

6.3 Ablation Study Results

Variant	Test Accuracy	Macro F1
Baseline (LR=0.0001)	0.9672	0.9653
Variant A (LR=0.001)	0.3089	0.1180
Variant B (Dropout=0.5)	0.9619	0.9596
Variant C (Fine-tune last 6)	0.9314	0.9277

An ablation study has been done to examine the effect of different training setups on the model performance based on the best performing architecture - VGG16. Several tests were carried out with varied learning rates, dropout ratios, and depths of fine-tuning.

It was found that the optimal combination is that of medium dropout and fine-tuning of the last layers of the convolutional neural network. Too much fine-tuning results in overfitting, while too little fine-tuning does not allow the model to learn MRI-specific features.

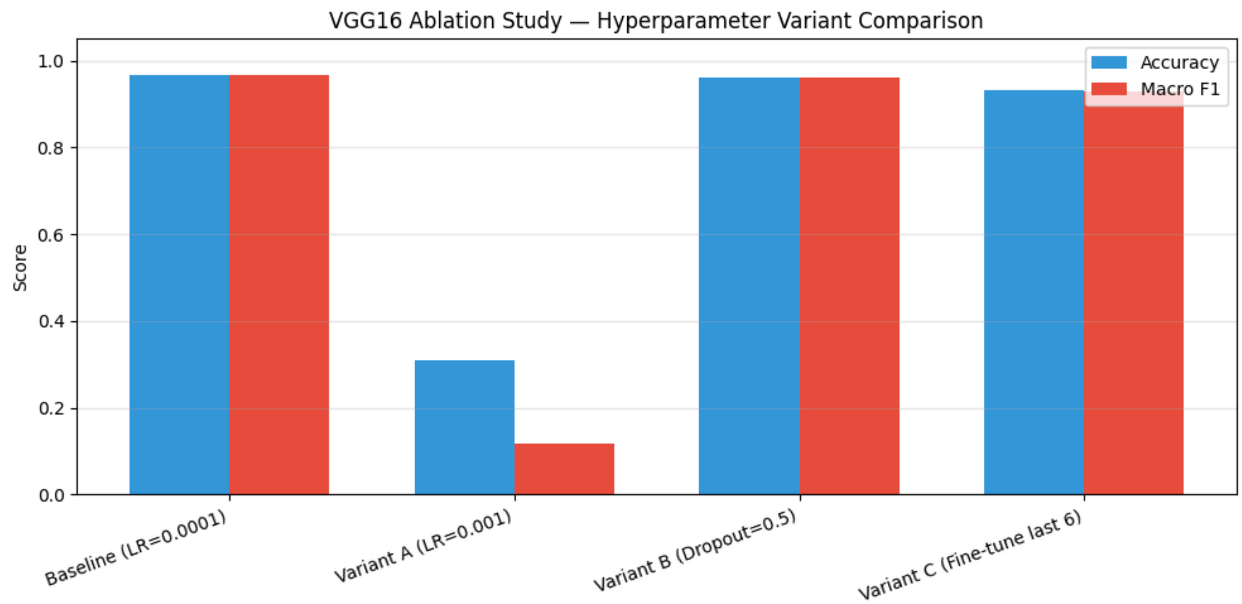
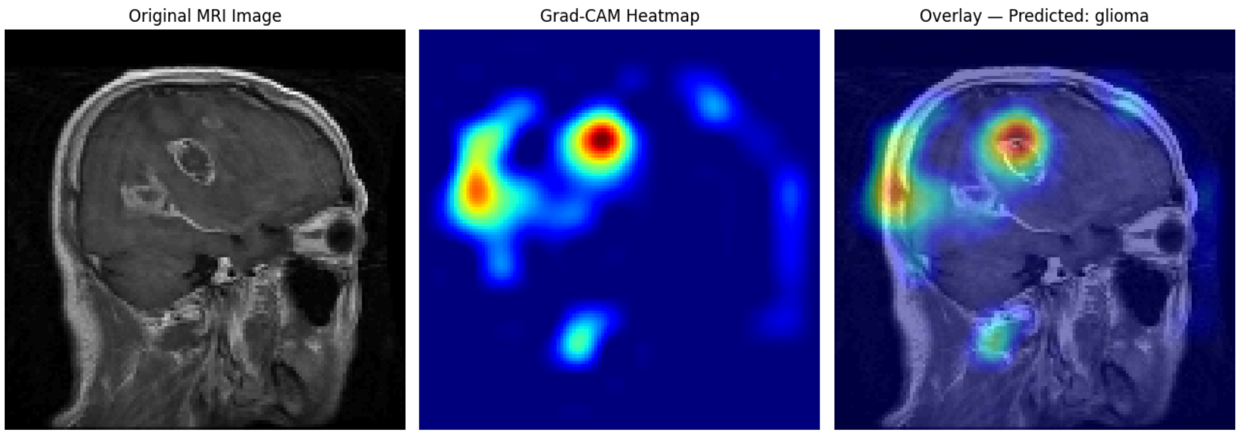


Figure 6.6: Comparative analysis of evaluation metrics across diverse VGG16 ablation study variants.

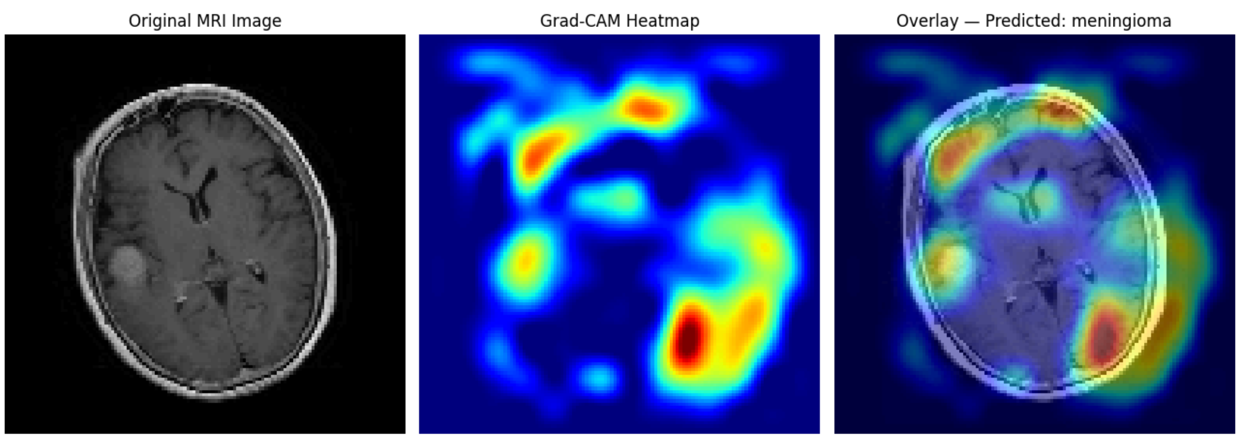
6.4 Explaining the Model Using Grad-CAM

To better understand the model and improve its transparency and explainability, Grad-CAM [6] was applied to MRI images that were correctly classified by the model. The resulting activation maps showed that the model concentrated mainly on the tumor regions and surrounding structures, rather than on background areas.

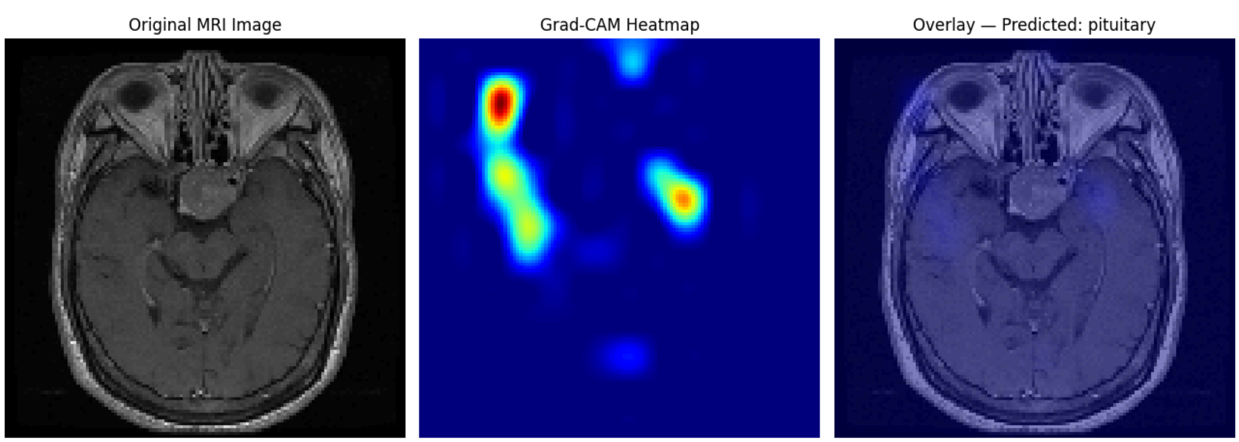
This provides further confidence in the correctness of the model's predictions. Although the model demonstrated high interpretability for glioma classification, the spatial localization for other tumor categories showed only a moderate correlation with the expected pathological regions.



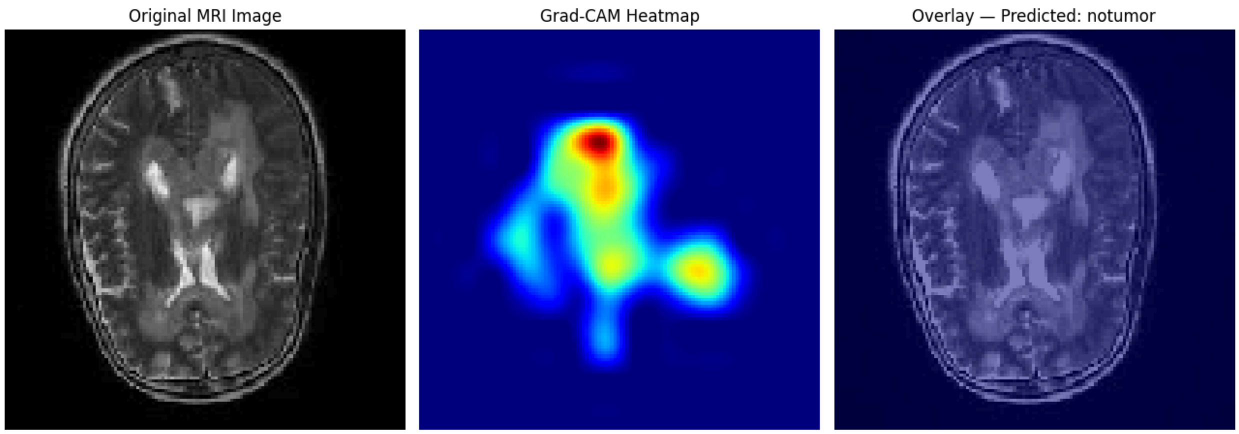
Grad-CAM for class: glioma



Grad-CAM for class: meningioma



Grad-CAM for class: pituitary



`Grad-CAM for class: notumor`

Figure 6.7: Representative Grad-CAM output illustrating the pathological focal points prioritized during model inference.

6.5 Discussion

The results of the experiments prove the high efficiency of using transfer learning-based convolutional networks for automatic brain tumor classification. Out of all the tested models, VGG16 showed the highest level of performance, which is indicative of the effectiveness of deep hierarchical features and transfer learning. MobileNetV2 gave the best balance between computational efficiency and accuracy of predictions, and thus could be considered one of the models suitable for usage in environments limited by computational resources. The U-Net demonstrated high performance due to its capability of preserving spatial information, whereas ResNet50 and CNN + LSTM showed low performance.

7. CONCLUSIONS

This project has provided a comparative analysis of five deep learning architectures for the problem of automated brain tumor classification by MRI images. Preprocessing of images included the application of a standardized pipeline for image resizing, normalization, label encoding, and data augmentation.

Based on the results obtained during testing of the five architectures on the considered dataset, the most successful architecture is VGG16 which demonstrated 96.72% of accuracy, Macro F1-Score 0.9653, and Weighted F1-Score 0.9670 due to effective transfer learning. In turn, the architecture MobileNetV2 showed the second best result (93.97%) but required much less computational power than the first one. At the same time, the encoder-based U-Net gave rather high results (90.85%), but ResNet50 and CNN + LSTM achieved quite low performance on the dataset used.

In conclusion, we can state that the results of experiments conducted show the potential of transfer learning-based convolutional neural networks as solutions to brain tumor classification from MRI images. Possible future research includes the usage of larger datasets, improved hyperparameters tuning, the use of attention-based architectures, and XAI approaches like Grad-CAM.

8. References

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